Immunomodulation and Anticancer Immunity: Reviewing the Potential of Probiotics and Their Delivery with Macromolecular Carriers

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ABSTRACT: Probiotics colonize in the gastrointestinal tract and regulate the homeostasis in healthy human hosts. These protect the host against putrefactive organisms and secrete soluble factors exhibiting important transductive roles. However, constitutive processes in human host are deregulated following dysbiosis caused during prolonged exposure to cytotoxic agents and pollutants. Apart from restoring the homeostasis, probiotic administration has shown to minimize carcinogenesis and post-surgery complications in cancer patients. Moreover, ability of microbial cells to colonize at tumor foci can be harnessed to deliver genes, therapeutic proteins and antibodies in a selective manner. In this review, we have discussed immunomodulatory roles of probiotics in context to cancer prevention. The article further proposes the use of dietary interventions for boosting anticancer immunity and as an alternative to detrimental chemotherapeutic agents. After summarizing clinical evidences on probiotic efficacy, formulation approaches have been described for effective delivery of the microorganisms. The literature shows that polysaccharide matrices can be employed to achieve the survivability of probiotics. Formulation approaches have been reviewed together with the risks associated with the migration of live microorganisms to systemic circulation and their ability of transmitting antibiotic resistance factors into human pathogens.

KEY WORDS: probiotics, cancer, immunomodulation, dysbiosis, polysaccharide matrix, biosafety

I. INTRODUCTION

Probiotics are live microorganisms, known to exist in symbiotic relationship inside human host.[1–3] In a healthy host, microorganisms remain separated from the intestinal epithelial cells with a protective mucus layer. Unlike pathogens which invade the barrier and assault the epithelial lining, probiotics remain localized and regulate the homeostasis of gastrointestinal (GI) tract.[4] In spite of showing species and strain-dependent variations, physiological effects of probiotics remain a popular subject in food and nutrition science. Evidences suggest that probiotic-rich food supplies the nutritional components and helps in combating the GI diseases. They protect the host by preventing the colonization of putrefactive and pathogenic organisms in the gut wall. Stains of *Lactobacillus* genera have shown other beneficial effects including secretion of hypocholesterolemic, anti-osteoporosis, anti-hypertensive and anti-inflammatory factors.[5–9] More specific roles of probiotics in human beings have been investigated over past two decades. For instance, it is now known that gut microbes regulate the response of T regulatory (Treg) cells and control the processes related to immune tolerance.[10] Consistent with these developments, number of probiotic products is increasing exponentially. Strains with well characterized physiological effects have been commercialized. Nonetheless, some of these have been used to produce pharmaceutically useful enzymes[11] and exopolysaccharides.[12–15]

Revelations on 'immunobiotic' effects have stimulated research on application of probiotics in anticancer therapeutics. Although microbes reside over the epithelial cells, their soluble factors diffuse through unstirred mucus layer and interact with the underlying cells.[16–18] At mucosal level, these detoxify ingested carcinogens, neutralize enteric antigens, produce inhibitory compounds and combat the metabolic activities related to carcinogenic transformations.[19–23] *Lactobacillus* and *Bifidobacillus* strains can physically bind to various carcinogens, such as heterocyclic aromatic amines, nitrosamines, aflatoxin B1, mycotoxins, polycyclic aromatic hydrocarbons, and phthalic acid esters. Mutagenic potential of these toxins is lowered as a result of adsorption over the microbial cells and subsequent metabolism.[24–26] At the same time, drop in pH during microbial production of short-chain fatty acids (acetate, propionate and butyrate) has been shown to retard the progression of colorectal cancers.[27,28] At systemic level, probiotics secrete soluble factors which participate in suppressing the pro-inflammatory cytokines.[29] These bind to extra-cellular receptors, and downregulate the events related to inflammation and uncontrolled cell proliferation (Fig. 1). Administration of probiotic VSL#3 in a rat model of colitis-associated cancer could indeed delay the transition from inflammation to dysplasia.[30] Altogether, the cascade contributes to initiation of adaptive response in the hosts. This has been verified through long-term (12 weeks or more) supplementation studies. Probiotics offered significant improvement in irritable bowel symptoms, mental health status and cancer-related fatigue in patients.[31,32]

Cross-talk between the microbiota and host is routinely challenged by physiological insults during prolonged exposure of antibiotics, food-borne chemicals and environmental pollutants.[33,34] Sequencing techniques have revealed that intestinal dysbiosis coincides with distortion in constitutive processes within the host. It alters the expression of microbial genes, especially of those involved in housekeeping metabolic events.[35,36] Recently, a study conducted in mice model has shown that ampicillin-induced dysbiosis creates imbalance in gut-brain axis (GBA). It caused deterioration in learning and memory of the animals.[37] GBA integrates peripheral intestinal functions with emotional and cognitive centers of the brain. This axis is maintained through bidirectional

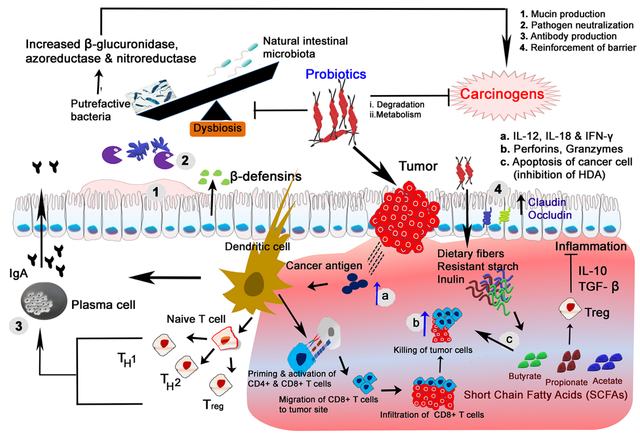

FIG. 1: Immunomodulatory role of probiotics in preventing cancer. At mucosal level, probiotic bacteria maintain the homeostasis by neutralizing putrefactive materials, producing inhibitory compounds and combating the metabolic activities related to carcinogenic transformations.[1–4] In addition, *Lactobacillus* and *Bifidobacillus* sp. can bind to and degrade the carcinogenic compounds, such as heterocyclic aromatic amines, nitrosamines, aflatoxin B1, mycotoxins, polycyclic aromatic hydrocarbons, and phthalic acid esters. At systemic level, probiotics contribute to generation of anti-cancer immunity. This is mediated through cross talk between bacterial metabolites, and immune and epithelial cells. In addition, probiotics can activate phagocytic cells for eliminating early-stage cancer cells. It occurs via activation of dendritic cells (DCs) which secrete increased levels of IL-12, IL-18 and IFN-γ. Moreover, DCs present the cancer antigens to Th cells in the lymph nodes which results into activation of antigen specific $CD8^+$ T cells. The Th1 cells secrete cytokines, the mediators of various anti-cancer effects (cytotoxicity, angiogenesis inhibition and antigen presentation). Later, $CD8^+$ T cells infiltrate into tumor tissue and exert their cytotoxic effect by producing granzymes, perforins and IFN-γ (a-c). Another mechanism of anticancer effect leans over production of short-chain fatty acids (SCFAs) through fermentation of dietary fibers (prebiotics). The SCFAs can activate Treg cells through binding to GPR43. It triggers the production of anti-inflammatory cytokines such as IL-10 and TGF-β. Also, probiotics regulate the production of anti-inflammatory cytokines which help in suppressing carcinogenesis.

neuroendocrine, immune and humoral communications.[38] Probiotics exhibit protective response over intestinal crypts and hippocampal neurons through reduction in oxidative stress.[37] Reversal of dysbiosis results into recovery in motor deficits, dopaminergic neuron loss and synaptic dysfunction.[39] In addition, scientists have elucidated profound role of probiotics in minimizing the post-surgical complications in cancer patients.

Seemingly, there is an emerging interest in harnessing the therapeutic potential of probiotics-based strategies for augmenting immunity against various types of cancer. Methods are being searched to improve the viability, biological property and stability of probiotics. In this review, we have described the role of probiotics in augmenting anticancer immunity. Formulation approaches for efficacious delivery of probiotics have also been discussed. The risks associated with the migration of microorganisms to systemic circulation and their ability of transmitting antibiotic resistance factors have also been reviewed.

II. PROBIOTICS AND IMMUNE ACTIVATION

Procarcinogenic role of microbiota has been illustrated through differences in the composition of tumor microbiome and that of adjacent non-neoplastic tissues. Whole genome sequencing on human tissue samples points toward enrichment of *Fusobacterium* sequences (*F. nucleatum*, *F. mortiferum*, and *F. necrophorum*) among colorectal cancer (CRC) patients.[40] Coupled with high virulence and adhesion, these can survive in hypoxic tumor environment and cause dysbiosis. *Fusobacterium* and its products promote cell proliferation and vasculature development within tumor microenvironment by stimulating tumor-promoting immune cell responses. It has been shown that proinflammatory microenvironment promotes the recruitment of tumor-infiltrating immune cells and supports progression of colorectal neoplasia even in the absence of colitis.[40,41] These effects are more powerful among individuals with pre-existing medical conditions.[32] For instance, patients with inflammatory bowel disease present about 10- to 40-fold higher risk of developing CRC. In such a case, chronically inflamed mucosa proceeds from dysplasia to high risk adenocarcinoma.[42] Therefore, scientists are exploring dietary approaches of restoring the balance in intestinal flora. The tactic is being explored as a means of suppressing the inflammation and carcinogenic events, locally as well as systemically. Some probiotics, such as *Bifidobacteria* and *Lactobacilli* ferment the dietary fibers and produce short-chain fatty acids (SCFAs) with important signaling roles.[43] For instance, butyrate induces apoptosis in cancer cells through histone deacetylase inhibition.[44,45] Significant correlation has been reported between progression of colorectal cancer and loss in SCFA-producing organisms. For instance, oral administration of *Butyrivibrio fibrisolvens* in mouse model has shown to reduce the number of colorectal aberrant crypt foci. It was attributed to: (1) increased numbers of natural killer (NK) and NKT cells, (2) decreased β-glucuronidase activity, and (3) butyrate production by the microorganisms.[46] Using interferon-gamma (IFN-γ) stimulated RAW 264.7 cells, it has been shown that butyrate represses the DNA binding and transcriptional activities of nuclear factor Kappa B, an important modulator of inflammation.[47]

Ohno's group[48–50] has demonstrated the anti-inflammatory and anti-apoptotic effects of acetate and butyrate at transcriptional level. Although butyrate induces the differentiation of regulatory T cells, acetate enhances the barrier function of intestinal epithelial cells. Germ-free mice died within 7 days when fed Shiga toxin (Stx) producing *E. coli* O157. Histological analysis of distal colon revealed lower number of goblet cells and signs of infiltration with inflammatory cells. However, mice survived when colonized with *Bifidobacterium longum* seven days before inoculation. In this case, serum concentration of Stx remained markedly lower and a positive correlation was noticed between the amount of fecal acetate and the resistance of animals to infection. Acetate prevented the reduction in transepithelial electrical resistance resulting from O157-induced cell death and translocation of Stx from apical to basolateral side of colonic epithelial cells.[49]

Later, the group illustrated the contribution of ABC-type carbohydrate transporter genes and acetate in preventing the infection (Fig. 2). Mice were fed with diet rich in acetylated starch, hydrolysable by intestinal bacteria. It offered a significant increase in fecal acetate, apparently discharged during the hydrolysis of starch. Interestingly, survival rate of starch-treated animals was similar to those administered with probiotics. On the contrary, those fed with nonprobiotic *Bifidobacteria* or carrying mutated transporters genes did not produce a sufficient amount of acetate. Apparently, preventive effects were not visible on these groups.[48]

Tumor development at extra-intestinal sites is checked through immunomodulatory effect of probiotics, as demonstrated by El-Nezami and coworkers.[51] About 40% reduction in hepatocellular tumor volume was attributed to down-regulated interleukin-17 (IL-17) secretion, and reduced population of proinflammatory and proangiogenic T helper 17 (Th17) cells. Authors suggested that probiotics limited the generation of excess Th17 cells in the gut which would otherwise have been recruited to perpetuate protumor inflammation in liver and other tissues. This was evident from significant reduction in Th17 population in the intestine and peripheral blood of treated animals. Microscopic analyses of excised tissues suggested that intra-tumoral vascularization and vessel sprouting was induced through hypoxia (Fig. 3). At the end of experiment (day 38), the difference of tumor weight/body weight between probiotic and cisplatin treated groups remained statistically insignificant.[51] In addition, commensals prime the tumor-associated innate myeloid cells for inflammatory cytokine production in response to concomitant immunotherapy. Investigations in germ-free mice revealed that disruption

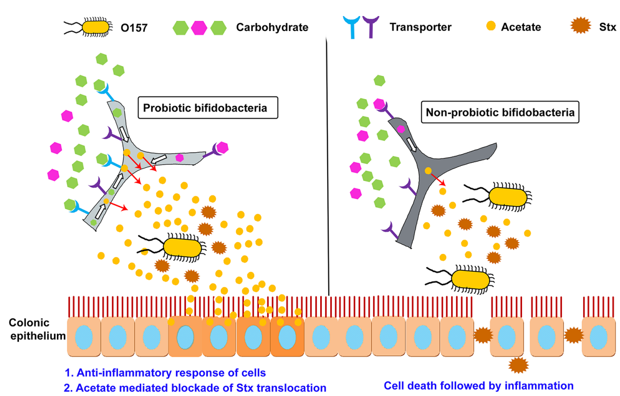

FIG. 2: Protective mechanism against O157 infection in mice by probiotic bifidobacteria. Nonprobiotic bifidobacteria-associated mice did not produce sufficient acetate molecules in the distal colon due to the lack of probiotic transporters. Epithelial cell death was induced with slight inflammation. Stx leaked into the blood and caused death of mice (Right). Mice associated with probiotic bifidobacteria produced sufficient amount of acetate in the presence of probiotic transporters by which sugars were actively taken in and metabolized. Acetate prevented the leakage of Stx into blood by reinforcing the barrier (Left). Modified from Fukuda et al.[48]

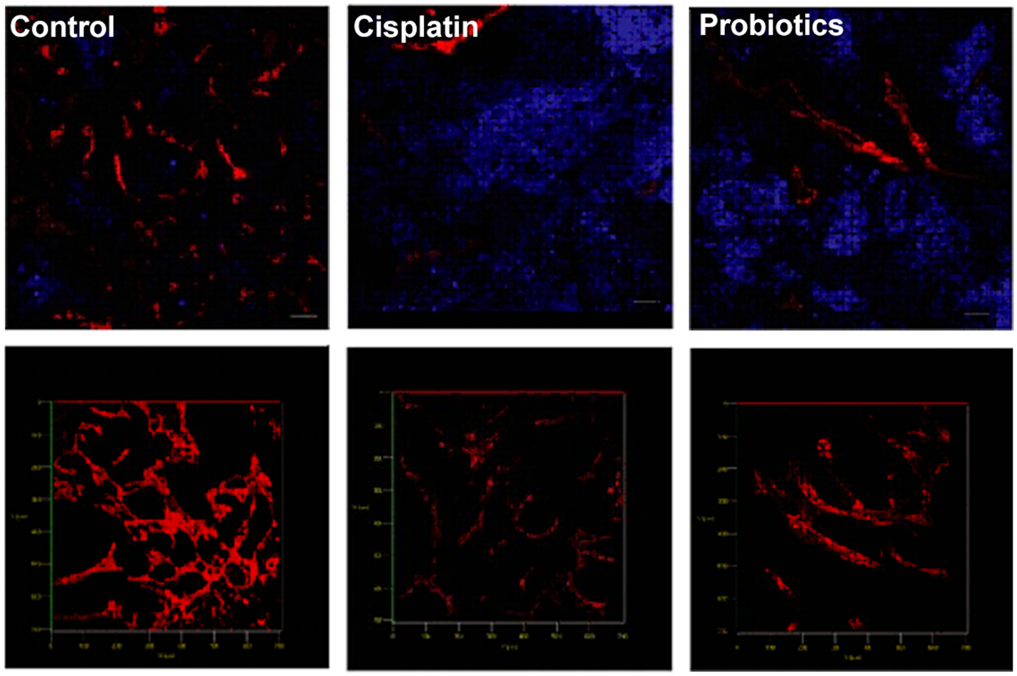

FIG. 3: Probiotics-mediated hypoxia in the tumor. Male C57BL6/N mice (n = 8) were fed with probiotics daily starting 1 week before subcutaneous injection of mouse hepatoma cell line. Upper panel shows confocal microscopic image of tumor sections immunostained for GLUT-1 (blue, hypoxic marker) and CD31 (red, angiogenesis marker). Lower panel shows optical sections (thickness, 25 μm) of respective image along Z-axis. The latter suggested that hypoxia was related to weakened angiogenesis. In comparison to control, lowering in microvessel density, relative vessel vascular area and number of vessel sprouts was recorded in probiotic treated animals (reprinted from Li et al.[51] with permission from PNAS, copyright 2016).

of microbiota impairs the early response of tumors to immuno- and chemo-therapy. Myeloid-derived cells responded poorly resulting in lower production of cytokine and tumor necrosis factor with immuno-treatment and deficient production of reactive oxygen species (ROS) and cytotoxicity after platinum therapy. The latter was verified from reduced expression of proinflammatory genes.[52]

Antitumor action of probiotics can also originate from the blockade of immunosuppressive cytokines, such as transforming growth factor (TGF-β) and IL-10. A combination of galunisertib (Gal, a TGF-β blocker) with *E. coli* strain Nissle induced more extensive necrosis within the tumor without causing acute loss in body weight. Stronger antitumor activity was evident from a powerful maturation of dendritic cells (DCs) and higher apoptosis in harvested tumor tissues. More importantly, Gal-probiotic combination produced significant reduction in the microvessel density and metastatic nodules in a xenograft mice model.[53] Such actions are sometimes augmented through enhanced bioavailability of ions and cofactors in the vicinity of tumor.[54,55]

Elimination of early-stage cancer cells leans over activation of phagocytic, T helper and cytotoxic T cells. The microorganisms interact with pattern recognition receptors in host cells, and prompt DCs to initiate potent antitumor response of Th1 and CD8+ T cells. Strains of *Lactobacilli* have been proven to suppress gastric cancer-related *H. pylori* infections in cultured human cells.[56] Treatment of DCs with *Lactobacillus* strains resulted into secretion of IL-12, IL-18, and IFN-γ and skewed the polarization of CD4+ and CD8+ T cells to Th1 and Tc1 cells. At this point, Th1 T cells secrete cytokines which mediate antigen presentation, cytotoxicity and angiogenesis inhibition.[57] In addition, these cells are involved in the priming of CD8+ cytotoxic T cells, the most efficient killers deployed by the immune system. This cascade can be potentiated with the administration of multiple microbes. For example, cocktail of three probiotics (*Bacillus mesentericus*, *Clostridium butyricum*, and *Enterococcus faecalis*) was reported to increase antigen presentation and activation markers on DCs and upregulate the production of IL-12.[58]

III. PREVENTION OF TUMORIGENESIS AND TUMOR-ASSOCIATED COMPLICATIONS

Therapeutic consequences of probiotic-cancer cell interactions have been studied with great interest. A number of probiotics have been investigated for cancer therapy in wild-type and genetically modified forms. Microbial cells colonize at primary tumor and systemic foci in response to signals from secreted chemokines. Their multiplication is supported by hypoxic tumor environment; more than 1000-fold higher proliferation has been recorded in comparison to healthy liver and kidney. Tumor-selective retention was further verifiable from their clearance among healthy animals without any evidence of organ deposition.[59–62] In these applications, immune system of host responds to incoming microbes by triggering the recruitment and activation of effector cells. This cascade is further potentiated by the residues discharged from the destruction of cancer cell and wall components of the microbe.[61]

Enzymatic battery of microbes can be exploited to simultaneously activate multiple prodrugs, as demonstrated by Tangney's group.[63] This can be accomplished through prior administration of vectors which accumulate in tumor bed and sensitize the cells against specific prodrug(s). Subsequently administered prodrugs are activated locally at these reservoirs and enable significant tumor regression. Balasari and coworkers[64] recorded significant reduction in metastases following targeted aerosolization of *Lactobacillus rhamnosus*. Probiotics contributed to overcoming the tolerance emerging from immunosuppressive commensals and promoting the maturation of resident antigen-presenting cells.[64]

Genetically engineered commensals from *Lactobacillus* and *Streptococcus* family have been used to deliver genes, therapeutic proteins and antibodies.[65,66] The technique enables expression of heterologous therapeutic agents, either by the bacterium itself or via plasmid transfer into host cells.[63,67] Transfected bacteria can home to tumors, replicate within them and locally express the loaded therapeutic proteins.[67] These proteins can induce apoptosis and inhibit neovascularization during proliferation of tumor cells. This was validated by Xia's group using *E. coli* Nissle 1917, loaded with p53 and Tum-5 proteins. Following injection in mice model, bacteria were detected in the tumor region within 72 h. As a result of this treatment, expression of Ki-67 was inhibited with a corresponding upregulation of caspase-3.[62] Very recently,[68] a probiotic bacterial system was developed for controlled production and intratumoral release of nanobodies, targeted at programmed cell death–ligand 1 (PD-L1) and cytotoxic T lymphocyte–associated protein-4 (CTLA-4). A single injection in mouse model showed enhanced therapeutic response in comparison to analogous clinically relevant antibodies. Most interestingly, no bacteria were detected in peripheral organs of animals even those showing completely regressed tumors. It underscores the ability of population to clear in sync with the disappearance of necrotic core. Therapeutic efficacy of such systems can be maximized by synchronizing the lysis of bacterial population at a threshold density. Prolonged intratumoral retention of bacterial vector minimizes the diffusion of payload to off-target sites and subsequent renal clearance.[69,70] However, biological application of genetically modified organisms has been challenged on the grounds of their possible environmental impact and socio-regulatory concerns.

A. Clinical Studies with Probiotics

Probiotics have been tested clinically under pre- and/or post-operative settings for prophylaxis and improvement in cancer-related complications (Table 1). In a pre-operative set up, these help preventing the development and subsequent invasion of enteropathogenic bacteria.[71,72] Probiotics are often supplemented with prebiotics, the non-digestible roughage derived from plants. The probiotic-prebiotic combination is termed synbiotics. Probiotics metabolize prebiotics into omega-3-fatty acids and other vital components.[73–75] Anti-inflammatory role of these molecules has been explained in earlier sections. Effectiveness of synbiotics has been demonstrated by Wilmes's group employing a microfluidics-based human microbial co-culture system. In comparison to individual components, symbiotic regimen caused downregulation of genes involved in procarcinogenic pathways. This results in reduced levels of lactate, an oncometabolite. Spheroid cultures treated with a molecular cocktail reflecting the synbiotic regimen exhibited attenuation in self-renewal capacity.[76] In addition, studies have shown that probiotics minimize post-operative infections and configure the gut ecosystem in manner to favor the recovery of surgical wounds. The former stems from the ability of probiotics to suppress the translocation of pathogens. This well contributed through neutralization of mutagens and stimulation of native commensals, as summarized in Fig. 1.

Positive correlation has been established between intestinal permeability of patients with probiotics administration and post-operative clinical variables.[84,87] In the presence of probiotics, host macrophages are stimulated to produce interleukins and trigger the differentiation of helper T cells. It later initiates the effector functions in an attempt to suppress oncogenic events.[79,81,85] In case of patients subjected to surgical interventions, these factors help in shortening the duration of post-operative hospitalization and the

TABLE 1: Clinical studies highlighting the role of probiotics in improving cancer related complications

Probiotic(s)	Study conditions	Important findings	Ref.
L. casei	Population - patients with superficial bladder cancer following transurethral resection (n = 138); study type –double blind, placebo controlled; groups - 3 [primary multiple tumors (A), recurrent single tumors (B), recurrent multiple tumors (C)]	Significant preventive effect on tumor recurrence with a hazard ratio of 2.58 (after exclusion of group C and correction for differences); downgradation of tumor from grade 2 to 1; less common upgrading in the treatment group	77
L. rhamnosus LC705 and *Propionibacterium freudenreichii* subsp. *Shermanii*	Population - healthy young men (n = 90); groups - 2; treatment schedule - 2 times/day for 5 weeks; control group - placebo	Significant decrease in the urinary concentration of aflatoxin B1-N7-guanine (reduced risk of liver cancer); reduction was 36 and 55%, respectively, at week 3 and 5	78
L. casei strain Shirota, *Bifidobacterium breve* strain Yakult and *galactooligosaccharides*	Population - patients with biliary cancer before hepatectomy (n = 101); groups - postoperative enteral feeding with synbiotics (A) and preoperative plus postoperative synbiotics (B)	Enhanced NK cell activity and reduced IL-6 concentration; higher postoperative concentrations of total organic acids and acetic acid in feces in group B than A ($P < 0.05$); postoperative infectious complications - 30.0% in group A and 12.1% in group B ($P < 0.05$)	79
Bifidobacterium longum (*B. longum*) and *L. johnsonii*	Population - subjects undergoing resection for colorectal cancer (n = 31); treatment - 2 doses/day for 3 days before operation and postoperatively from day 2 to day 4; control group - placebo	No effect on DC phenotype or activation; *ex vivo* stimulation with lipopolysaccharides showed lower proliferation rate; patients colonized with *Lactobacillus* showed significantly lower expression of CD83-123, CD83-HLADR, and CD83-11c	80
L. plantarum, *L. acidophilus* and *B. longum*	Population - patients (n = 100) with colorectal carcinoma divided into treatment (n = 50) and control (maltodextrin, n = 50) groups; treatment schedule - orally for 6 days preoperatively and 10 days postoperatively	Increased transepithelial resistance ($P < 0.05$), reduced bacterial translocation ($P < 0.05$); enhanced mucosal tight junction protein expression improved post-operative recovery of peristalsis, improvement in infection-related complication and gut defecation functions	81

TABLE 1: (*continued*)

Probiotic(s)	Study conditions	Important findings	Ref.
B. longum, *L. acidophilus*, and *Enterococcus faecalis*	Population - patients with colorectal cancer (n = 60); treatment -3 days before surgery; thrice a day	Lowered levels of endotoxins, D-lactic acids, serum IL-6 and C-reactive protein; overall postoperative infectious complications (10.0% versus 33.3% in placebo, $P < 0.05$)	82
L. casei strain Shirota	Population - women with an HPV + low-grade squamous intraepithelial lesion (n = 54); treatment – daily drink up to 6 months	Two-fold improvement in the clearance of HPV-infection-related cytological abnormality in treatment group; higher HPV clearance rates	83
L. plantarum, *L. acidophilus* and *B. longum*	Population - patients with colorectal carcinoma screened for absence of distant metastasis (n = 150); treatment - 6 days preoperatively and 10 days postoperatively	Lower infection rate (reduced serum zonulin, a marker of septicemia) in the probiotics group ($P < 0.05$); reduction in the duration of postoperative pyrexia and antibiotic therapy; inhibition in p38 mitogen-activated protein kinase signaling pathway	84
B. longum, *L. acidophilus* and *Enterococcus faecalis*	Population - patients diagnosed with confined CRC (n = 60); treatment - probiotic (n = 30) or placebo (n = 30); treatment - 5 days before and 7 days after CRC resection	Significant recovery of bowel function upon perioperative probiotic administration; improvement in days to first flatus (P = 0.03) and the days to first defecation (P = 0.03); reduced incidence of diarrhea (P = 0.0352)	85
L. acidophilus, *L. lactis*, *L. casei*, *B. longum*, *B. bifidum* and *B. infantis*	population - CRC patients (n = 52); treatment – after four weeks of surgery twice daily for six months with the product containing 30 billion colony-forming units	Significant reduction in the level of pro-inflammatory cytokines; no post-surgical infections occurred, and no antibiotics were required	86

HPV, human papillomavirus.

period needed for antibiotics administration.[88] In a study on CRC patients, a significant reduction in antibiotic usage and hospital stay length ($P < 0.001$) was observed following 7 days of probiotic intake prior to surgery.[71] Revelations on metabolic cross-feeding suggest that regimens must be selected to enhance competition between the gut microbiome and host cells.[76] It is worthwhile to mention that limited availability of dietary substrates is likely to check the proliferation of cancerous cells.

Probiotics can minimize the population of *Fusobacterium*, a genus suspected for its role in tumorigenesis within the intestine. A single center study on CRC patients has shown that the population of mucosa-associated pathogens can be minimized significantly through perioperative normalization of dysbiosis. Probiotic treated patients showed about six-fold reduction in the relative abundance of *Fusobacteria*. This interpretation was based on pyrosequencing of V3 region of the bacterial 16S rRNA gene in tissue samples collected from the tumor sites during surgery.[89] Interventions leaning over restoration of normal microbiota can be helpful in delaying the tumor growth and/or minimizing the relapses arising from miniscule tumor deposits left undetected during surgical resection.

Efficacy of probiotics-based preoperative interventions has been questioned by some research groups. It is argued that probiotics either produce placebo level effects or inflict serious adverse effects. In a pilot study of McNaught et al.,[90] *L. plantarum* did not offer significant benefits in terms of gastric colonization, pathogen translocation and postoperative septic morbidity. Instead, single IV administration of antibiotic prevented postoperative infections more effectively among colon cancer patients.[88] A report of Dutch regulatory authorities highlighted infectious complications among probiotic-treated acute pancreatitis patients. In a placebo-controlled trial, two-fold higher mortality in the probiotics group was attributed to its deleterious effects on the small bowel wall. This was verified from the inflammatory signs in the bowel wall of autopsy samples of patients with bowel ischemia. Authors argued that metabolic stress was aggravated with incremental probiotic load and accompanying oxygen demand.[91]

B. Biosafety of Probiotics

Concerns over the use of probiotics arise from their ability to alter the barrier properties of intestinal epithelium. Also, these may migrate to vascular and extra-vascular compartments, and transfer the plasmids carrying antibiotic resistance or pathogenic genes. Chances of invasion through the intestinal membrane are higher among very young and immuno-compromised patients, and those with pre-existing inflammation in gut wall. This allows migration of microbes or their virulence factors to blood, internal organs, and otherwise sterile tissues such as the mesenteric lymph nodes.[81] Bacteremia and sepsis have been reported with *Lactobacillus* species.[92] A recent trial showed inverse association between probiotic administration in breastfeeding mothers and immunomodulatory effects in infants.[10] Microbes were transported from the mother to infant via enteromammary pathway. This transport loop was created by DCs which ferried the microbes through tight junction of intestinal epithelium and drained into the mammary gland via lymphatic system. Changes in the colonization pattern of infants manifested as high frequency of enteric and respiratory infections in first two-years of life. Nonetheless, such adverse effects have so far not been reported in healthy adult subjects even following translocation through gut wall.[93]

Looking at above reports, it is discernable that therapeutic application of probiotics must have oversight of possible risks related to mobile genetic elements. Considering the high frequency of transfer, probiotics can spread their antibiotic resistance factors to intestinal pathogens and other commensals.[94] This may allow the transmission of antibiotic resistance determinants into human pathogens via food chain. For example, tetracycline and erythromycin resistance genes were indeed transferred from two wild-type *L. plantarum* strains to *Enterococcus faecalis* in germ-free rats. Animals were inoculated with donor strains after one week of inoculating with the recipient. Transconjugants (TCs), detected in fecal samples, were comparable for the two donors.[95] Considering the consumption of probiotics on regular basis, these events are expected to multiply with TCs acting as donor, particularly among vulnerable patients. This may lead to emergence of the multiple-drug resistance strains and the consequences have been addressed in earlier reviews.[96,97]

IV. FORMULATION DEVELOPMENT OF PROBIOTICS

The International Dairy Federation recommends that active microbes should not be less than 1×10^7 colony-forming units/g in order to achieve successful therapeutic effects.[98,99] However, viability of cells is potentially challenged during practical use conditions. It is typical to traditional practice of administering the organisms admixed in ready-to-eat dairy products. In addition, it is argued that poor survivability emanates from cold and osmotic stress during storage of dairy products. As a result, commercial products exhibit low tolerance against bowel cleansing, changes in pH and action of digestive enzymes during transit along the GI tract.[100–102] For instance, *Bifidobacteria* show low survivability in gastric juice and/or exposure to oxygen.[103] Eventually, these factors contribute to sub-optimal colonization in the target region.

A plausible alternative relies over extraction of active principles from corresponding suspension culture.[104,105] Metabolites (< 10 kDa fraction) derived from *L. plantarum* reduced the proliferation of melanoma cells and secretion of pro-inflammatory cytokines in H4/TLT co-culture model.[106] Secreted and surface proteins from *Lactobacillus* and *Bifidobacterium* strains have also shown regulatory effects.[105] Nevertheless, commercial interest in these molecules is limited because of complex isolation and identification procedures.

These limitations have been addressed by applying long standing principles of formulation design. Such formulations are expected to offer stealth characteristics to the microorganisms within harsh gut environment.[107] These must have design features appropriate to evade antimicrobial defenses and immunological surveillance after consumption of the product.[108] Accordingly, present day probiotic formulation research focuses on improving the encapsulation efficiency, colonization and shelf life of the microorganisms. Alginate (ALG) based matrices have been frequently used as carrier, particularly for developing multiparticulate preparations. Properties of formulations can be improved through chemical modification or mixing with other polysaccharides or proteins. The formulator takes into account detrimental effects of unfavorable temperature, humidity and redox conditions during formulation processing.[54,109] At the same time, cells must be protected against dehydration and stress during secondary processes, such as lyophilization and spray drying.[107,110] Formulations must be manufactured in compliance with good manufacturing practices while qualifying the standards for safety and efficacy as applicable to drug delivery systems. Sustainability of individual approach is assessed on the basis of revival and discharge of loaded cells.

A. Hydrogel

Hydrogels are often three-dimensional crosslinked networks developed from native or derivatized polymers. These can retain large amount of water or biological fluid without dissolving, and are attractive from the standpoint of biocompatibility, tissue-like softness and tunable porosity.[111] Optimum combination of hydrocolloids and other additives may be employed to develop hybrid hydrogels exhibiting superior properties.[112] Although polymer offers structural support, additives exhibit specific biological functions in hydrogels for tissue engineering and wound healing applications.[113] Gel matrix can be reinforced with additional components in order to impart stimuli-responsiveness, resistance against enzymatic digestion and controlled erosion kinetics. Hydrogels can form diverse physical structures, including beads and other multi-particulates. Structural developments are controlled during molecular assemblage of aggregating units in aqueous medium.[114–116] Advancements in chemical modification and controlled radical polymerization techniques have led to production of tunable microporous hydrogels exhibiting optimized mechanical strength, rheological behavior and, anti-fatigue characteristics under biorelevant hydrodynamic conditions. One such ingestible hydrogel pill could withstand repeated mechanical loads in the stomach of a pig model for up to 1 month.[117] The formulation consisted of superabsorbent hydrogel particles which allowed quick water imbibition and retention within a hydrogel membrane.

Hydrogel preparations are developed from food grade polymers (often polysaccharides or their derivatives). This includes microgels developed from regenerated cellulose derivatives showing interpenetrated nanofibrils and high surface area. Porous structures offer high loading efficiency as these allow surface adherence and penetration of microbial cells into the core of microgels. Cells can be protected against gastric secretions and bile salts by enclosing the microgels within acid-resistant calcium alginate hydrogel. Release of cells from such a core-shell composite has been found proportional to the matrix porosity.[118] The discharge of microbes can further be controlled with the use of rigidized core or multi-layered shells. Ionic interaction between oppositely charged polymers has been employed to produce a mechanically strong matrix. The technique avoids the use of toxic chemical crosslinking agents. Resistance of the hydrogel against dissolution in the upper GI tract permits colonization of loaded microbes in the distal part of intestine.[119]

B. Microcapsules and Beads

Microcapsules are the most extensively studied formulations for the encapsulation of probiotics. These are largely developed from polysaccharides obtainable from plant, animal and microbial sources. Polysaccharides are attractive because of natural origin and availability of chemical handles which can be utilized for derivatization and cross-linking.[116,120,121] Microcapsules can be prepared through spray-drying, polyelectrolyte complexation, emulsification and coacervation. The multiparticulate nature of the formulation offers large specific area for encapsulation of the microorganisms. In addition, its core-shell structure allows co-loading with other functional molecules, such as omega-3 fatty acids. Fatty acids protect the microorganisms in acidic environment and facilitate their colonization in the intestinal wall. The former is attributable to limited diffusion of H^+ ions and oxygen in the presence of lipids. Importantly, spray and freeze-dried microcapsules offer high shelf life because of low water activity.[122,123]

Another mechanism of protection relies on coating the microcapsules with pH-sensitive polymers, as demonstrated previously.[124,125] ALG microcapsules remained stable at gastric pH (fasted and fed) when coated with oppositely charged chitosan (CHI). Microbes were discharged at intestinal pH following the dissociation of CHI molecules ($pK_a \sim 6.5$) from ALG surface. As a result, number of released viable cells were significantly higher in coated microcapsules in comparison those from uncoated microcapsules. The number of layers can be increased to achieve desirable level of gastric pH resistance and survival during dry storage. Individual layer can be tuned at nanoscopic level with the use of layer-by-layer deposition.[126] The technique allows sequential deposition of alternating layers of cationic and anionic polyelectrolytes over the microorganism. Strong electrostatic interactions offer a strong barrier even with the use of minimum concentration of polymers. In addition to pH resistance, microbes encapsulated with three CHI/ALG layers could maintain their ability to grow and proliferate.[127]

Some groups have used amino acid, protein and surfactants as coating material with the objective of improving the release rate. Surfactants are commercially available in a range of hydrophobicity and molecular characteristics. Depending upon chemical structure, their interaction with the core is contributed by charge as well as hydrophobicity. Accordingly, the type and concentration of surfactant can be optimized for improving the micromeritic and mechanical properties of microcapsules. Ionic surfactants cause a loss in microbial population despite their superior interaction and penetration into oppositely charged cores. This is attributable to anti-microbial activity of ionic surfactants. On the contrary, lecithin-coated ALG cores exhibited complete recovery of *L. plantarum* after 1 h exposure to simulated gastric fluid and 2 h in simulated intestinal fluid.[128] Damage to the cells from heat and dehydration is controlled through incorporation of proteins, such as soy protein isolate, casein and whey protein isolate. Here, proteins coronate the microbial cells via electrostatic forces. Owing to high glass transition temperature, proteins are transformed to glassy state during temperature changes which in turn retards the molecular mobility within the cytoplasm.[129] The reader is referred to earlier reviews on the use of microencapsulation approaches for probiotics delivery.[130,131]

C. Nanofibers

Use of nanofibers in the delivery of nutraceuticals and probiotics is an emerging area of research. Monolithic and composite nanofibers are developed through electrospinning of polymeric dispersion seeded with the microorganisms. In uniaxial mode, the dispersion is forced electrostatically through a spinneret tip. Potential difference between the tip and collector surface whips and stretches the discharge into continuous fibers of nanosized diameter in a single step.[110,132,133] Rod shaped organisms and their spores are aligned longitudinally along the length of fiber.[134] Low radius of nanofibers reduces the radiative heat flux and thus, offers better survivability to the microorganisms during thermal processing.[135] Core-shell nanofibers can be produced by carrying out the electrospinning process in coaxial mode. Here, the spinneret consists of two concentric needles, and the rate of flow of core and shell forming dispersion is controlled by separate syringe pumps (Fig. 4). A comparison between uniaxial and double-layered fibers suggests that the latter offer superior protection to encapsulated cells against biorelevant temperature and pH alterations.[136]

L. plantarum has shown high survivability in fructo-oligosaccharide nanofibers even under high voltage electric field and moist heat (70°C) conditions.[137] Recently, Berlec and coworkers[138] demonstrated successful incorporation of *Lactobacillus* species into polyethylene oxide nanofibers, irrespective of size of the microorganisms (74–10.82 μm). However, species with hydrophobic cell surface showed superior survivability during electrospinning in comparison to counterparts with hydrophilic surface. For

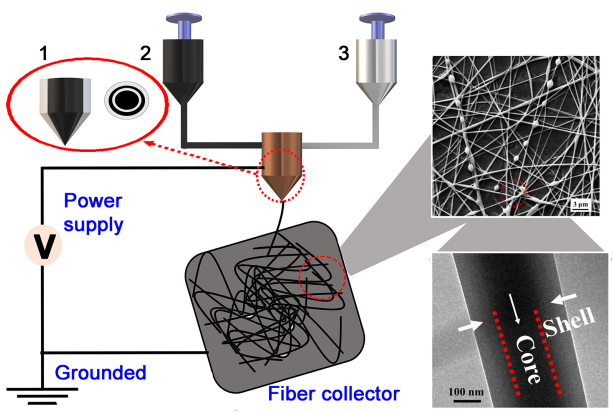

FIG. 4: One-step coaxial electrospinning set-up for the encapsulation of probiotics. The nanofibers show beaded morphology and core-shell structure (reprinted from Feng et al.[136] with permission from Elsevier, copyright 2020).

example, in comparison to 2.8 log reduction in the viability of hydrophilic *L. bulgaris*, negligible loss occurred during encapsulation of hydrophobic *L. gasseri*.[138]

Aspect ratio, porosity and morphological properties of nanofibers can be tuned to accommodate large population of active probiotics. Superior stability of nanofibers has been recorded with the use of plasticizing agents. Its mechanistic aspects have been investigated employing glycerol as plasticizer in starch formate (SF) nanofibers. Plasticization effect of glycerol was regulated by its interaction at core-sheath interface depending upon temperature. At room temperature, SF-water interactions dominated because of strong H-bonding ability of the oxygen atom. However, weakening of H bonds at elevated temperature allowed progressive diffusion of glycerol between SF chains. As a result of plasticization, glass transition temperature of SF dropped from 205 to 40°C.[139] The thermal stability of nanofibers can be exploited to develop scaffolds capable of discharging the microbes for longer period of time. Similarly, probiotic-loaded composite mats are now being employed in bioactive and antimicrobial food packages. Probiotics protect the preparation against spoilage via production of antimicrobial metabolites or competing with pathogenic organisms.[140]

Apart from above-described formulation approaches, immunotherapeutic response may be amplified through the administration of purified strains. Tested administration models include placing the microbial dispersion at the tumor bed or administering through oral and intravenous (IV) routes. Oral administration is convenient and works well for solid tumors located in the gut wall. However, translocation of microorganisms to extra-intestinal tumor locations is impeded by the host's defense. This apparently leads to marginal reduction in the bioavailability of orally administered microbes, particularly aiming at distant tumor sites.[67] Parenteral routes are more effective in such cases. Superiority of IV route ($P < 0.001$) was recently demonstrated using *Bifidobacterium bifidum* in human papillomavirus-induced tumor model. Potent antitumor response was verified from the activation of tumor-specific IL-12 and IFN-γ, lymphocyte proliferation, and CD8+ cytolytic response in the spleen.[141] This explains the use of IV route in majority of investigations carried out on animal models.

V. CONCLUSION AND FUTURE DIRECTIONS

Clinical studies have established that probiotics prevent a number of opportunistic infections by interfering with pathogen colonization.[83,142] However, interpretations on the efficacy of probiotics based anticancer therapeutics must be drawn with caution. Given the diversity of microbiota and their variable colonization, clinical outcomes may vary among the strains. Lack of effectiveness can sometimes emerge from shorter administration schedule, formulation composition, poor survivability of organisms during oral administration and complexity of surgical procedures.[81] Sometimes, observations may be confounded by the factors, such as small sample size, lack of randomization, protocol violations and small follow-up period. Considering these points and limited sample size of clinical studies, much research remains to be carried out before integrating probiotics with modern therapeutic modalities.

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REFERENCES

1. Lenoir-Wijnkoop IGL, Roy D, Reid G. The clinical and economic impact of probiotics consumption on respiratory tract infections: Projections for Canada. PLoS One. 2016;11:e0166232.
2. Härtel C, Pagel J, Spiegler J, Buma J, Henneke P, Zemlin M, Viemann D, Gille C, Gehring S, Frommhold D, Rupp J. Lactobacillus acidophilus/Bifidobacterium infantis probiotics are associated with increased growth of VLBWI among those exposed to antibiotics. Sci Rep. 2017;7:5633.
3. Akash MS, Fiayyaz F, Rehman K, Sabir S, Rasool MH. Gut microbiota and metabolic disorders: Advances in therapeutic interventions. Crit Rev Immunol. 2019;39(4):223–7.
4. Zafar H, Jain AG, Khetpal N, Ali S, Rashid MU, Ahmad S. Small intestinal bacterial overgrowth: A critical review of an underrecognized but disrupting entity. Crit Rev Oncog. 2020;25(4):365–79.
5. Mikelsaar M, Zilmer M. Lactobacillus fermentum ME-3 – an antimicrobial and antioxidative probiotic. Microb Ecol Health Dis. 2009;21:1–27.
6. Chiang SS, Pan TM. Beneficial effects of Lactobacillus paracasei subsp. paracasei NTU 101 and its fermented products. Appl Microbiol Biotechnol. 2012;93:903–16.
7. Archer AC, Muthukumar SP, Halami PM. Anti-inflammatory potential of probiotic Lactobacillus spp. on carrageenan induced paw edema in Wistar rats. Int J Biol Macromol. 2015;81:530–7.
8. Rondanelli M, Faliva MA, Perna S, Giacosa A, Peroni G, Castellazzi AM. Using probiotics in clinical practice: Where are we now? A review of existing meta-analyses. Gut Microbes. 2017;8: 521–43.
9. Markowiak P, Śliżewska K. Effects of probiotics, prebiotics, and synbiotics on human health. Nutrients. 2017;9:1021.
10. Quin C, Estaki M, Vollman DM, Barnett JA, Gill SK, Gibson DL. Probiotic supplementation and associated infant gut microbiome and health: A cautionary retrospective clinical comparison. Sci Rep. 2018;8(1):8283.
11. Gandhi D, Chanalia P, Attri P, Dhanda S. Dipeptidyl peptidase-II from probiotic Pediococcus acidilactici: Purification and functional characterization. Int J Biol Macromol. 2016;93:919–32.
12. Pompei A, Cordisco L, Amaretti A, Zanoni S, Matteuzzi D, Rossi M. Folate production by bifidobacteria as a potential probiotic property. Appl Environ Microbiol. 2007;73:179–85.
13. Abid Y, Casillo A, Gharsallah H, Joulak I, Lanzetta R, Corsaro MM, Attia H, Azabou S. Production and structural characterization of exopolysaccharides from newly isolated probiotic lactic acid bacteria. Int J Biol Macromol. 2018;108:718–28.
14. Ayyash M, Abu-Jdayil B, Itsaranuwat P, Galiwango E, Tamiello-Rosa C, Abdullah H, Esposito G, Hunashal Y, Obaid RS, Hamed F. Characterization, bioactivities, and rheological properties of exopolysaccharide produced by novel probiotic Lactobacillus plantarum C70 isolated from camel milk. Int J Biol Macromol. 2020;144:938–46.
15. Junter GA, Karakasyan C. Polysaccharides against viruses: Immunostimulatory properties and the delivery of antiviral vaccines and drugs. Crit Rev Ther Drug Carrier Syst. 2020;37(1):1–64.
16. Atarashi K, Tanoue T, Ando M, Kamada N, Nagano Y, Narushima S, Suda W, Imaoka A, Setoyama H, Nagamori T, Ishikawa E. Th17 cell induction by adhesion of microbes to intestinal epithelial cells. Cell. 2015;163(2):367–80.
17. Diaz-Garrido N, Fábrega MJ, Vera R, Giménez R, Badia J, Baldomà L. Membrane vesicles from the probiotic Nissle 1917 and gut resident Escherichia coli strains distinctly modulate human dendritic cells and subsequent T cell responses. J Funct Foods. 2019;61:103495.

18. Santos AM, Dos Santos CC, de Oliveira TB, Serafini MR, Alves IA. What's new in allergen immunotherapy patents? A review. Crit Rev Ther Drug Carrier Syst. 2021;38(6):105–25.
19. Lee J, Yun HS, Cho KW, Oh S, Kim SH, Chun T, Bongjoon K, Whang KY. Evaluation of probiotic characteristics of newly isolated Lactobacillus spp.: Immune modulation and longevity. Int J Food Microbiol. 2011;148(2):80–6.
20. Cousin FJ, Jouan-Lanhouet S, Dimanche-Boitrel MT, Corcos L, Jan G. Milk fermented by Propionibacterium freudenreichii induces apoptosis of HGT-1 human gastric cancer cells. PLoS One. 2012;7(3):e31892.
21. Lee NK, Son SH, Jeon EB, Jung GH, Lee JY, Paik HD. The prophylactic effect of probiotic Bacillus polyfermenticus KU3 against cancer cells. J Funct Foods. 2015;14:513–8.
22. Aziz N, Bonavida B. Activation of natural killer cells by probiotics. For Immunopathol Dis Therap. 2016;7(1-2):41–55.
23. Kulwal V, Baxi K, Sawarkar SP, Bhatt LK. Colorectal cancer management by herbal drug-based nanocarriers: An overview. Crit Rev Ther Drug Carrier Syst. 2020;37(1):65–104.
24. Nowak A, Libudzisz Z. Ability of probiotic Lactobacillus casei DN 114001 to bind or/and metabolise heterocyclic aromatic amines in vitro. Eur J Nutr 2009;48:419–27.
25. Faridnia F, Hussin A, Saari N, Mustafa S, Yee L, Manap M. In vitro binding of mutagenic heterocyclic aromatic amines by Bifidobacterium pseudocatenulatum G4. Benef Microbes. 2010;1:149–54.
26. Nowak A, Kuberski S, Libudzisz Z. Probiotic lactic acid bacteria detoxify N-nitrosodimethylamine. Food Addit Contam Part A. 2014;31:1678–87.
27. Worthley DL, Le Leu RK, Whitehall VL, Conlon M, Christophersen C, Belobrajdic D, Mallitt KA, Hu Y, Irahara N, Ogino S, Leggett BA. A human, double-blind, placebo-controlled, crossover trial of prebiotic, probiotic, and synbiotic supplementation: Effects on luminal, inflammatory, epigenetic, and epithelial biomarkers of colorectal cancer. Am J Clin Nutr. 2009;90(3):578–86.
28. Huang L, Shan YJ, He CX, Ren MH, Tian PJ, Song W. Effects of L. paracasei subp. paracasei X12 on cell cycle of colon cancer HT-29 cells and regulation of mTOR signalling pathway. J Funct Foods. 2016;21:431–9.
29. Mi H, Dong Y, Zhang B, Wang H, Peter CC, Gao P, Fu H, Gao Y. Bifidobacterium infantis ameliorates chemotherapy-induced intestinal mucositis via regulating T cell immunity in colorectal cancer rats. Cell Physiol Biochem. 2017;42:2330–41.
30. Appleyard CB, Cruz ML, Isidro AA, Arthur JC, Jobin C, De Simone C. Pretreatment with the probiotic VSL#3 delays transition from inflammation to dysplasia in a rat model of colitis-associated cancer. Am J Physiol Liver Physiol. 2011;301:G1004–13.
31. Ohigashi S, Hoshino Y, Ohde S, Onodera H. Functional outcome, quality of life, and efficacy of probiotics in postoperative patients with colorectal cancer. Surg Today. 2011;41(9):1200–6.
32. Lee JY, Chu SH, Jeon JY, Lee MK, Park JH, Lee DC, Lee JW, Kim NK. Effects of 12 weeks of probiotic supplementation on quality of life in colorectal cancer survivors: A double-blind, randomized, placebo-controlled trial. Dig Liver Dis. 2014;46(12):1126–32.
33. Defois C, Ratel J, Garrait G, Denis S, Le Goff O, Talvas J, Mosoni P, Engel E, Peyret P. Food chemicals disrupt human gut microbiota activity and impact intestinal homeostasis as revealed by in vitro systems. Sci Rep. 2018;8(1):11006.
34. Tiwari S. Mannosylated constructs as a platform for cell-specific delivery of bioactive agents. Crit Rev Ther Drug Carrier Syst. 2018;35(2):157–94.
35. Fulbright LE, Ellermann M, Arthur JC. The microbiome and the hallmarks of cancer. PLoS Pathog. 2017;13(9):e1006480.
36. Pellegrini M, Ippolito M, Monge T, Violi R, Cappello P, Ferrocino I, Cocolin LS, De Francesco A, Bo S, Finocchiaro C. Gut microbiota composition after diet and probiotics in overweight breast cancer survivors: A randomized open-label pilot intervention trial. Nutrition. 2020;74:110749.
37. Sarkar SR, Mazumder PM, Banerjee S. Probiotics protect against gut dysbiosis associated decline in learning and memory. J Neuroimmunol. 2020;348:577390.

38. Carabotti M, Scirocco A, Maselli MA, Severi C. The gut-brain axis: Interactions between enteric microbiota, central and enteric nervous systems. Ann Gastroenterol. 2015;28(2):203–9.
39. Sun J, Li H, Jin Y, Yu J, Mao S, Su KP, Ling Z, Liu J. Probiotic Clostridium butyricum ameliorated motor deficits in a mouse model of Parkinson's disease via gut microbiota-GLP-1 pathway. Brain Behav Immun. 2021;91:703–15.
40. Kostic AD, Gevers D, Pedamallu CS, Michaud M, Duke F, Earl AM, Ojesina AI, Jung J, Bass AJ, Tabernero J, Baselga J. Genomic analysis identifies association of Fusobacterium with colorectal carcinoma. Genome Res. 2012;22(2):292–8.
41. Kostic AD, Chun E, Robertson L, Glickman JN, Gallini CA, Michaud M, Clancy TE, Chung DC, Lochhead P, Hold GL, El-Omar EM. Fusobacterium nucleatum potentiates intestinal tumorigenesis and modulates the tumor-immune microenvironment. Cell Host Microbe. 2013;14(2):207–15.
42. Saito Y, Hinoi T, Adachi T, Miguchi M, Niitsu H, Kochi M, Sada H, Sotomaru Y, Sakamoto N, Sentani K, Oue N. Synbiotics suppress colitis-induced tumorigenesis in a colon-specific cancer mouse model. PLoS One. 2019;14(6):e0216393.
43. Zhu H, Cao C, Wu Z, Zhang H, Sun Z, Wang M, Xu H, Zhao Z, Wang Y, Pei G, Yang Q. The probiotic L. casei Zhang slows the progression of acute and chronic kidney disease. Cell Metab. 2021;33(10):2091–3.
44. Wei W, Sun W, Yu S, Yang Y, Ai L. Butyrate production from high-fiber diet protects against lymphoma tumor. Leuk Lymphoma. 2016;57:2401–8.
45. LeBlanc JG, Chain F, Martín R, Bermúdez-Humarán LG, Courau S, Langella P. Beneficial effects on host energy metabolism of short-chain fatty acids and vitamins produced by commensal and probiotic bacteria. Microb Cell Fact. 2017;16:79.
46. Ohkawara S, Furuya H, Nagashima K, Asanuma N, Hino T. Oral administration of Butyrivibrio fibrisolvens, a butyrate-producing bacterium, decreases the formation of aberrant crypt foci in the colon and rectum of mice. J Nutr. 2005;135:2878–83.
47. Park JS, Lee EJ, Lee JC, Kim WK, Kim HS. Anti-inflammatory effects of short chain fatty acids in IFN-γ-stimulated RAW 264.7 murine macrophage cells: Involvement of NF-κB and ERK signaling pathways. Int Immunopharmacol. 2007;7:70–7.
48. Fukuda S, Toh H, Taylor TD, Ohno H, Hattori M. Acetate-producing bifidobacteria protect the host from enteropathogenic infection via carbohydrate transporters. Gut Microbes. 2012;3(5):449–54.
49. Fukuda S, Toh H, Hase K, Oshima K, Nakanishi Y, Yoshimura K, Tobe T, Clarke JM, Topping DL, Suzuki T, Taylor TD. Bifidobacteria can protect from enteropathogenic infection through production of acetate. Nature. 2011;469(7331):543–7.
50. Furusawa Y, Obata Y, Fukuda S, Endo TA, Nakato G, Takahashi D, Nakanishi Y, Uetake C, Kato K, Kato T, Takahashi M. Commensal microbe-derived butyrate induces the differentiation of colonic regulatory T cells. Nature. 2013;504(7480):446–50.
51. Li J, Sung CY, Lee N, Ni Y, Pihlajamäki J, Panagiotou G, El-Nezami H. Probiotics modulated gut microbiota suppresses hepatocellular carcinoma growth in mice. Proc Natl Acad Sci U S A. 2016;113(9):E1306–15.
52. Iida N, Dzutsev A, Stewart CA, Smith L, Bouladoux N, Weingarten RA, Molina DA, Salcedo R, Back T, Cramer S, Dai RM. Commensal bacteria control cancer response to therapy by modulating the tumor microenvironment. Science. 2013;342(6161):967–70.
53. Shi L, Sheng J, Wang M, Luo H, Zhu J, Zhang B, Liu Z, Yang X. Combination therapy of TGF-beta blockade and commensal-derived probiotics provides enhanced antitumor immune response and tumor suppression. Theranostics. 2019;9(14):4115–29.
54. Dafe A, Etemadi H, Dilmaghani A, Mahdavinia GR. Investigation of pectin/starch hydrogel as a carrier for oral delivery of probiotic bacteria. Int J Biol Macromol. 2017;97:536–43.
55. Li Y, Feng C, Li J, Mu Y, Liu Y, Kong M, Cheng X, Chen X. Construction of multilayer alginate hydrogel beads for oral delivery of probiotics cells. Int J Biol Macromol. 2017;105(1):924–30.
56. Chen X, Liu XM, Tian F, Zhang Q, Zhang HP, Zhang H, Chen W. Antagonistic activities of

Lactobacilli against Helicobacter pylori growth and infection in human gastric epithelial cells. J Food Sci. 2012;77:9–14.
57. Iwasaki A, Medzhitov R. Toll-like receptor control of the adaptive immune responses. Nat Immunol. 2004;5:987–95.
58. Hua MC, Lin TY, Lai MW, Kong MS, Chang HJ, Chen CC. Probiotic Bio-Three induces Th1 and anti-inflammatory effects in PBMC and dendritic cells. World J Gastroenterol. 2010;16:3529–40.
59. Stritzker J, Weibel S, Hill PJ, Oelschlaeger TA, Goebel W, Szalay AA. Tumor-specific colonization, tissue distribution, and gene induction by probiotic Escherichia coli Nissle 1917 in live mice. Int J Med Microbiol. 2007;297:151–62
60. Danino T, Lo J, Prindle A, Hasty J, Bhatia SN. In vivo gene expression dynamics of tumor-targeted bacteria. ACS Synth Biol. 2012;1(10):465–70.
61. Zheng JH, Nguyen VH, Jiang SN, Park SH, Tan W, Hong SH, Shin MG, Chung IJ, Hong Y, Bom HS, Choy HE. Two-step enhanced cancer immunotherapy with engineered Salmonella typhimurium secreting heterologous flagellin. Sci Transl Med. 2017;9(376):eaak9537.
62. He L, Yang H, Tang J, Liu Z, Chen Y, Lu B, He H, Tang S, Sun Y, Liu F, Ding X. Intestinal probiotics E. coli Nissle 1917 as a targeted vehicle for delivery of p53 and Tum-5 to solid tumors for cancer therapy. J Biol Eng. 2019;13:58.
63. Lehouritis P, Stanton M, McCarthy FO, Jeavons M, Tangney M. Activation of multiple chemotherapeutic prodrugs by the natural enzymolome of tumour-localised probiotic bacteria. J Control Release. 2016;222:9–17.
64. Le Noci V, Guglielmetti S, Arioli S, Camisaschi C, Bianchi F, Sommariva M, Storti C, Triulzi T, Castelli C, Balsari A, Tagliabue E. Modulation of pulmonary microbiota by antibiotic or probiotic aerosol therapy: A strategy to promote immunosurveillance against lung metastases. Cell Rep. 2018;24(13):3528–38.
65. Steidler L, Neirynck S, Huyghebaert N, Snoeck V, Vermeire A, Goddeeris B, Cox E, Remon JP, Remaut E. Biological containment of genetically modified Lactococcus lactis for intestinal delivery of human interleukin 10. Nat Biotechnol. 2003;21(7):785–9.
66. Robert S, Steidler L. Recombinant Lactococcus lactis can make the difference in antigen-specific immune tolerance induction, the type 1 diabetes case. Microb Cell Fact. 2014;13(Suppl 1):S11.
67. Cronin M, Morrissey D, Rajendran S, El Mashad SM, Van Sinderen D, O'sullivan GC, Tangney M. Orally administered bifidobacteria as vehicles for delivery of agents to systemic tumors. Mol Ther. 2010;18(7):1397–407.
68. Gurbatri CR, Lia I, Vincent R, Coker C, Castro S, Treuting PM, Hinchliffe TE, Arpaia N, Danino T. Engineered probiotics for local tumor delivery of checkpoint blockade nanobodies. Sci Transl Med. 2020;12(530):eaax0876.
69. Pawelek JM, Low KB, Bermudes D. Tumor-targeted Salmonella as a novel anticancer vector. Cancer Res. 1997;57(20):4537–44.
70. Zhang L, Gao L, Zhao L, Guo B, Ji K, Tian Y, Wang J, Yu H, Hu J, Kalvakolanu DV, Kopecko DJ. Intratumoral delivery and suppression of prostate tumor growth by attenuated Salmonella enterica serovar typhimurium carrying plasmid-based small interfering RNAs. Cancer Res. 2007;67(12):5859–64.
71. Polakowski CB, Kato M, Preti VB, Schieferdecker MEM, Ligocki Campos AC. Impact of the preoperative use of synbiotics in colorectal cancer patients: A prospective, randomized, double-blind, placebo-controlled study. Nutrition. 2019;58:40–6.
72. Tanaka K, Yano M, Motoori M, Kishi K, Miyashiro I, Ohue M, Ohigashi H, Asahara T, Nomoto K, Ishikawa O. Impact of perioperative administration of synbiotics in patients with esophageal cancer undergoing esophagectomy: A prospective randomized controlled trial. Surgery. 2012;152(5):832–42.
73. Ishikawa H, Akedo I, Otani T, Suzuki T, Nakamura T, Takeyama I, Ishiguro S, Miyaoka E, Sobue T, Kakizoe T. Randomized trial of dietary fiber and Lactobacillus casei administration for prevention of colorectal tumors. Int J Cancer. 2005;116(5):762–7.
74. Rayes N, Seehofer D, Theruvath T, Mogl M, Langrehr JM, Nüssler NC, Bengmark S, Neuhaus P.

Effect of enteral nutrition and synbiotics on bacterial infection rates after pylorus-preserving pancreatoduodenectomy: A randomized, double-blind trial. Ann Surg. 2007;246(1):36–41.
75. Arun KB, Madhavan A, Reshmitha TR, Thomas S, Nisha P. Short chain fatty acids enriched fermentation metabolites of soluble dietary fibre from Musa paradisiaca drives HT29 colon cancer cells to apoptosis. PLoS One. 2019;14(5):e0216604.
76. Greenhalgh K, Ramiro-Garcia J, Heinken A, Ullmann P, Bintener T, Pacheco MP, Baginska J, Shah P, Frachet A, Halder R, Fritz JV. Integrated in vitro and in silico modeling delineates the molecular effects of a synbiotic regimen on colorectal-cancer-derived cells. Cell Rep. 2019;27(5):1621–32.e9.
77. Aso Y, Akaza H, Kotake T, Tsukamoto T, Imai K, Naito S. Preventive effect of a Lactobacillus casei preparation on the recurrence of superficial bladder cancer in a double-blind trial. The BLP Study Group. Eur Urol. 1995;27(2):104–9.
78. El-Nezami HS, Polychronaki NN, Ma J, Zhu H, Ling W, Salminen EK, Juvonen RO, Salminen SJ, Poussa T, Mykkänen HM. Probiotic supplementation reduces a biomarker for increased risk of liver cancer in young men from Southern China. Am J Clin Nutr. 2006;83(5):1199–203.
79. Sugawara G, Nagino M, Nishio H, Ebata T, Takagi K, Asahara T, Nomoto K, Nimura Y. Perioperative synbiotic treatment to prevent postoperative infectious complications in biliary cancer surgery: A randomized controlled trial. Ann Surg. 2006;244(5):706–14.
80. Gianotti L, Morelli L, Galbiati F, Rocchetti S, Coppola S, Beneduce A, Gilardini C, Zonenschain D, Nespoli A, Braga M. A randomized double-blind trial on perioperative administration of probiotics in colorectal cancer patients. World J Gastroenterol. 2010;16(2):167–75.
81. Liu Z, Qin H, Yang Z, Xia Y, Liu W, Yang J, Jiang Y, Zhang H, Yang Z, Wang Y, Zheng Q. Randomised clinical trial: The effects of perioperative probiotic treatment on barrier function and post-operative infectious complications in colorectal cancer surgery - a double-blind study. Aliment Pharmacol Ther. 2011;33(1):50–63.
82. Zhang JW, Du P, Gao J, Yang BR, Fang WJ, Ying CM. Preoperative probiotics decrease postoperative infectious complications of colorectal cancer. Am J Med Sci. 2012;343(3):199–205.
83. Verhoeven V, Renard N, Makar A, Van Royen P, Bogers JP, Lardon F, Peeters M, Baay M. Probiotics enhance the clearance of human papillomavirus-related cervical lesions: A prospective controlled pilot study. Eur J Cancer Prev. 2013;22(1):46–51.
84. Liu ZH, Huang MJ, Zhang XW, Wang L, Huang NQ, Peng H, Lan P, Peng JS, Yang Z, Xia Y, Liu WJ. The effects of perioperative probiotic treatment on serum zonulin concentration and subsequent postoperative infectious complications after colorectal cancer surgery: A double-center and double-blind randomized clinical trial. Am J Clin Nutr. 2013;97(1):117–26.
85. Yang Y, Xia Y, Chen H, Hong L, Feng J, Yang J, Yang Z, Shi C, Wu W, Gao R, Wei Q. The effect of perioperative probiotics treatment for colorectal cancer: Short-term outcomes of a randomized controlled trial. Oncotarget. 2016;7(7):8432–40.
86. Zaharuddin L, Mokhtar NM, Nawawi KN, Ali RA. A randomized double-blind placebo controlled trial of probiotics in post-surgical colorectal cancer. BMC Gastroenterology. 2019;19:131.
87. Collins MD, Gibson GR. Probiotics, prebiotics, and synbiotics: Approaches for modulating the microbial ecology of the gut. Am J Clin Nutr. 1999;69(5):1052S–7S.
88. Sadahiro S, Suzuki T, Tanaka A, Okada K, Kamata H, Ozaki T, Koga Y. Comparison between oral antibiotics and probiotics as bowel preparation for elective colon cancer surgery to prevent infection: Prospective randomized trial. Surgery. 2014;155:493–503.
89. Gao Z, Guo B, Gao R, Zhu Q, Wu W, Qin H. Probiotics modify human intestinal mucosa-associated microbiota in patients with colorectal cancer. Mol Med Rep. 2015;12(4):6119–27.
90. McNaught CE, Woodcock NP, MacFie J, Mitchell CJ. A prospective randomised study of the probiotic Lactobacillus plantarum 299V on indices of gut barrier function in elective surgical patients. Gut. 2002;51(6):827–31.
91. Besselink MG, van Santvoort HC, Buskens E, Boermeester MA, van Goor H, Timmerman HM, Nieuwenhuijs VB, Bollen TL, van Ramshorst B, Witteman BJ, Rosman C. Probiotic prophylaxis

in predicted severe acute pancreatitis: A randomised, double-blind, placebo-controlled trial. Lancet. 2008;371(9613):651–9.
92. Yan F, Cao H, Cover TL, Washington MK, Shi Y, Liu L, Chaturvedi R, Peek RM, Wilson KT, Polk DB. Colon-specific delivery of a probiotic-derived soluble protein ameliorates intestinal inflammation in mice through an EGFR-dependent mechanism. J Clin Invest. 2011;121(6):2242–53.
93. Liong MT. Safety of probiotics: Translocation and infection. Nutr Rev. 2008;66(4):192– 202.
94. Gevers D, Huys G, Swings J. In vitro conjugal transfer of tetracycline resistance from Lactobacillus isolates to other Gram-positive bacteria. FEMS Microbiol Lett. 2003;225(1):125–30.
95. Jacobsen L, Wilcks A, Hammer K, Huys G, Gevers D, Andersen SR. Horizontal transfer of tet(M) and erm(B) resistance plasmids from food strains of Lactobacillus plantarum to Enterococcus faecalis JH2-2 in the gastrointestinal tract of gnotobiotic rats. FEMS Microbiol Ecol. 2007;59(1): 158–66.
96. Zheng M, Zhang R, Tian X, Zhou X, Pan X, Wong A. Assessing the risk of probiotic dietary supplements in the context of antibiotic resistance. Front Microbiol. 2017;8:908.
97. Das DJ, Shankar A, Johnson JB, Thomas S. Critical insights into antibiotic resistance transferability in probiotic Lactobacillus. Nutrition. 2020;69:110567.
98. Song H, Yu W, Gao M, Liu X, Ma X. Microencapsulated probiotics using emulsification technique coupled with internal or external gelation process. Carbohydr Polym. 2013;96(1):181–9.
99. Zahra D, Javaid A, Iqbal M, Akbar I, Ashfaq UA. Synthesis and therapeutic potential of nanoceria against cancer: An update. Crit Rev Ther Drug Carrier Syst. 2021;38(5):1–26.
100. Muthukumarasamy P, Allan-Wojtas P, Holley RA. Stability of Lactobacillus reuteri in different types of microcapsules. J Food Sci. 2006;71:20–4.
101. Pourjafar H, Noori N, Gandomi H, Basti AA, Ansari F. Viability of microencapsulated and non-microencapsulated Lactobacilli in a commercial beverage. Biotechnol Rep. 2020;25:e00432.
102. Chang Y, Yang Y, Xu N, Mu H, Zhang H, Duan J. Improved viability of Akkermansia muciniphila by encapsulation in spray dried succinate-grafted alginate doped with epigallocatechin-3-gallate. Int J Biol Macromol. 2020;159:373–82.
103. Su R, Zhu XL, Fan DD, Mi Y, Yang CY, Jia X. Encapsulation of probiotic Bifidobacterium longum BIOMA 5920 with alginate–human-like collagen and evaluation of survival in simulated gastrointestinal conditions. Int J Biol Macromol. 2011;49:979–84.
104. Sabikhi L, Babu R, Thompkinson DK, Kapila S. Resistance of microencapsulated Lactobacillus acidophilus LA1 to processing treatments and simulated gut conditions. Food Bioprocess Tech 2010;3:586–93.
105. Yan F, Polk DB. Characterization of a probiotic-derived soluble protein which reveals a mechanism of preventive and treatment effects of probiotics on intestinal inflammatory diseases. Gut Microbes. 2012;3(1):25–8.
106. Dimitrovski D, Cencič A, Winkelhausen E, Langerholc T. Lactobacillus plantarum extracellular metabolites: In vitro assessment of probiotic effects on normal and cancerogenic human cells. Int Dairy J. 2014;39:293–300.
107. Mirtic J, Rijavec T, Zupancic S, Zvonar Pobirk A, Lapanje A, Kristl J. Development of probiotic-loaded microcapsules for local delivery: Physical properties, cell release and growth. Eur J Pharm Sci. 2018;121:178–87.
108. Dodoo CC, Wang J, Basit AW, Stapleton P, Gaisford S. Targeted delivery of probiotics to enhance gastrointestinal stability and intestinal colonisation. Int J Pharm. 2017;530(1-2):224–9.
109. Ghibaudo F, Gerbino E, Copello GJ, Dall'Orto VC, Gómez-Zavaglia A. Pectin-decorated magnetite nanoparticles as both iron delivery systems and protective matrices for probiotic bacteria. Colloids Surf B Biointerfaces. 2019;180:193–201.
110. Skrlec K, Zupancic S, Prpar Mihevc S, Kocbek P, Kristl J, Berlec A. Development of electrospun nanofibers that enable high loading and long-term viability of probiotics. Eur J Pharm Biopharm. 2019;136:108–19.

111. Yucel Falco C, Falkman P, Risbo J, Cardenas M, Medronho B. Chitosan-dextran sulfate hydrogels as a potential carrier for probiotics. Carbohydr Polym. 2017;172:175–83.
112. Alehosseini A, del Pulgar EM, Fabra MJ, Gómez-Mascaraque LG, Benítez-Páez A, Sarabi-Jamab M, Ghorani B, Lopez-Rubio A. Agarose-based freeze-dried capsules prepared by the oil-induced biphasic hydrogel particle formation approach for the protection of sensitive probiotic bacteria. Food Hydrocolloids. 2019;87:487–96.
113. Phan VG, Duong HT, Thambi T, Nguyen TL, Turabee MH, Yin Y, Kim SH, Kim J, Jeong JH, Lee DS. Modularly engineered injectable hybrid hydrogels based on protein-polymer network as potent immunologic adjuvant in vivo. Biomaterials. 2019;195:100–10.
114. Argin S, Kofinas P, Lo YM. The cell release kinetics and the swelling behavior of physically crosslinked xanthanechitosan hydrogels in simulated gastrointestinal conditions. Food Hydrocolloids. 2014;40:138–44.
115. Du X, Zhou J, Shi J, Xu B. Supramolecular hydrogelators and hydrogels: From soft matter to molecular biomaterials. Chem Rev. 2015;115(24):13165–307.
116. Tiwari S, Patil R, Dubey SK, Bahadur P. Derivatization approaches and applications of pullulan. Adv Colloid Interface Sci. 2019;269:296–308.
117. Liu X, Steiger C, Lin S, Parada GA, Liu J, Chan HF, Yuk H, Phan NV, Collins J, Tamang S, Traverso G. Ingestible hydrogel device. Nat Commun. 2019;10(1):493.
118. Li W, Zhu Y, Ye F, Li B, Luo X, Liu S. Probiotics in cellulose houses: Enhanced viability and targeted delivery of Lactobacillus plantarum. Food Hydrocolloids. 2017;62:66–72.
119. Falco CY, Falkman P, Risbo J, Cárdenas M, Medronho B. Chitosan-dextran sulfate hydrogels as a potential carrier for probiotics. Carbohydr Polym. 2017;172:175–83.
120. Tiwari S, Patil R, Bahadur P. Polysaccharide based scaffolds for soft tissue engineering applications. Polymers. 2018;11(1):1.
121. Wen R, Umeano AC, Chen P, Farooqi AA. Polymer-based drug delivery systems for cancer. Crit Rev Ther Drug Carrier Syst. 2018;35(6) 521–53.
122. Eratte D, McKnight S, Gengenbach TR, Dowling K, Barrow CJ, Adhikari BP. Co-encapsulation and characterisation of omega-3 fatty acids and probiotic bacteria in whey protein isolate–gum Arabic complex coacervates. J Funct Foods. 2015;19:882–92.
123. Eratte D, Wang B, Dowling K, Barrow CJ, Adhikari B. Survival and fermentation activity of probiotic bacteria and oxidative stability of omega-3 oil in co-microcapsules during storage. J Funct Foods. 2016;23:485–96.
124. Cook MT, Tzortzis G, Charalampopoulos D, Khutoryanskiy VV. Production and evaluation of dry alginate-chitosan microcapsules as an enteric delivery vehicle for probiotic bacteria. Biomacromolecules. 2011;12(7):2834–40.
125. Jose C, Amra K, Bhavsar C, Momin M, Omri A. Polymeric lipid hybrid nanoparticles: Properties and therapeutic applications. Crit Rev Ther Drug Carrier Syst. 2018;35(6) 555–88.
126. Wang M, Yang J, Li M, Wang Y, Wu H, Xiong L, Sun Q. Enhanced viability of layer-by-layer encapsulated Lactobacillus pentosus using chitosan and sodium phytate. Food Chem. 2019;285:260–5.
127. Anselmo AC, McHugh KJ, Webster J, Langer R, Jaklenec A. Layer-by-layer encapsulation of probiotics for delivery to the microbiome. Adv Mater. 2016;28(43):9486–90.
128. Albadran HA, Chatzifragkou A, Khutoryanskiy VV, Charalampopoulos D. Development of surfactant-coated alginate capsules containing Lactobacillus plantarum. Food Hydrocolloids. 2018;82:490–9.
129. Gong P, Di W, Yi H, Sun J, Zhang L, Han X. Improved viability of spray-dried Lactobacillus bulgaricus sp1.1 embedded in acidic-basic proteins treated with transglutaminase. Food Chem. 2019;281: 204–12.
130. Cook MT, Tzortzis G, Charalampopoulos D, Khutoryanskiy VV. Microencapsulation of probiotics for gastrointestinal delivery. J Control Release. 2012;162(1):56–67.
131. Seifert A, Kashi Y, Livney YD. Delivery to the gut microbiota: A rapidly proliferating research field. Adv Colloid Interface Sci. 2019;274:102038.

132. Fung WY, Yuen KH, Liong MT. Agrowaste-based nanofibers as a probiotic encapsulant: Fabrication and characterization. J Agric Food Chem. 2011;59(15):8140–7.
133. Sharma R, Singh H, Joshi M, Sharma A, Garg T, Goyal AK, Rath G. Recent advances in polymeric electrospun nanofibers for drug delivery. Crit Rev Ther Drug Carrier Syst. 2014;31(3):187–217.
134. Zupancic S, Rijavec T, Lapanje A, Petelin M, Kristl J, Kocbek P. Nanofibers with incorporated autochthonous bacteria as potential probiotics for local treatment of periodontal disease. Biomacromolecules. 2018;19(11):4299–306.
135. Wu H, Fan J, Qin X, Zhang G. Thermal radiative properties of electrospun superfine fibrous PVA films. Mater Lett. 2008;62:828–31.
136. Feng K HR, Wu R, Wei Y, Zong M, Linhardt RJ, Wu H. A novel route for double-layered encapsulation of probiotics with improved viability under adverse conditions. Food Chem. 2020;310:125977.
137. Feng K, Zhai MY, Zhang Y, Linhardt RJ, Zong MH, Li L, Wu H. Improved viability and thermal stability of the probiotics encapsulated in a novel electrospun fiber mat. J Agric Food Chem. 2018;66(41):10890–7.
138. Zupancic S, Skrlec K, Kocbek P, Kristl J, Berlec A. Effects of electrospinning on the viability of ten species of lactic acid bacteria in poly(Ethylene oxide) nanofibers. Pharmaceutics. 2019;11(9):483.
139. Lancuski A, Abu Ammar A, Avrahami R, Vilensky R, Vasilyev G, Zussman E. Design of starch-formate compound fibers as encapsulation platform for biotherapeutics. Carbohydr Polym. 2017;158:68–76.
140. Zabihollahi N, Alizadeh A, Almasi H, Hanifian S, Hamishekar H. Development and characterization of carboxymethyl cellulose based probiotic nanocomposite film containing cellulose nanofiber and inulin for chicken fillet shelf life extension. Int J Biol Macromol. 2020;160:409–17.
141. Abdolalipour E, Mahooti M, Salehzadeh A, Torabi A, Mohebbi SR, Gorji A, Ghaemi A. Evaluation of the antitumor immune responses of probiotic Bifidobacterium bifidum in human papillomavirus-induced tumor model. Microb Pathog. 2020;145:104207.
142. Spurbeck RR, Arvidson CG. Lactobacilli at the front line of defense against vaginally acquired infections. Future Microbiol. 2011;6(5):567–82.